

EC466-966: Estimation and Inference in Econometrics

Exercise # 1- Answer Key

1. Please see EC466hw1sol.pdf.
2. Show that $\mathbf{X}'\mathbf{X}$ is a symmetric positive definite matrix.

We first show that $\mathbf{X}'\mathbf{X}$ is a symmetric matrix. Let $\mathbf{A} = \mathbf{X}'\mathbf{X}$, then

$$\mathbf{A}' = (\mathbf{X}'\mathbf{X})' = \mathbf{X}'\mathbf{X} = \mathbf{A},$$

therefore $\mathbf{X}'\mathbf{X}$ is a symmetric matrix.

To show that $\mathbf{X}'\mathbf{X}$ is a positive definite matrix, first note that

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]'$$

By the definition of a positive definite matrix, then we take a $K \times 1$ non-zero vector $\mathbf{z}$ and calculate $\mathbf{z}'\mathbf{X}'\mathbf{X}\mathbf{z}$.

$$\mathbf{z}'\mathbf{X}'\mathbf{X}\mathbf{z} = \sum_{i=1}^n \mathbf{z}'\mathbf{x}_i\mathbf{x}_i'\mathbf{z} = \sum_{i=1}^n (\mathbf{z}'\mathbf{x}_i)^2 > 0,$$

since $\mathbf{z}'\mathbf{x}_i$ is a scalar, i.e., $\mathbf{z}'\mathbf{x}_i = \mathbf{x}_i'\mathbf{z}$.

3. Show that $(\mathbf{X}'\mathbf{X})^{-1}$ is a symmetric positive definite matrix.

To show that $(\mathbf{X}'\mathbf{X})^{-1}$ is a symmetric positive definite matrix, we use the eigendecomposition of a symmetric matrix. Since we have shown that $\mathbf{X}'\mathbf{X}$ is a symmetric positive definite matrix, we can write that

$$\mathbf{X}'\mathbf{X} = \mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}',$$

where $\mathbf{\Gamma}$ is an orthogonal matrix (i.e., $\mathbf{\Gamma}\mathbf{\Gamma}' = \mathbf{I}_K$) whose columns are the eigenvectors of $\mathbf{X}'\mathbf{X}$, and $\mathbf{\Lambda}$ is a diagonal matrix whose entries are the eigenvalues of $\mathbf{X}'\mathbf{X}$. Then,

$$(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}')^{-1} = (\mathbf{\Gamma}')^{-1}\mathbf{\Lambda}^{-1}\mathbf{\Gamma}^{-1} = \mathbf{\Gamma}\mathbf{\Lambda}^{-1}\mathbf{\Gamma}',$$

since $\mathbf{\Gamma}' = \mathbf{\Gamma}^{-1}$. Note that all eigenvalues of a positive definite matrix is positive, i.e., $\lambda_i > 0$ for all $i = 1, 2, \dots, k$ where λ_i is the eigenvalues of $\mathbf{X}'\mathbf{X}$. $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_K)$. Then $\mathbf{\Lambda}^{-1} = \text{diag}(1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_K)$. Therefore, the eigenvalues of $(\mathbf{X}'\mathbf{X})^{-1}$ is also positive. This implies that $(\mathbf{X}'\mathbf{X})^{-1}$ is a positive definite matrix.

Now, we need to show that $(\mathbf{X}'\mathbf{X})^{-1}$ is a symmetric matrix. N

$$((\mathbf{X}'\mathbf{X})^{-1})' = (\mathbf{\Gamma}\mathbf{\Lambda}^{-1}\mathbf{\Gamma}')' = (\mathbf{\Gamma}')'(\mathbf{\Lambda}^{-1})'\mathbf{\Gamma}' = \mathbf{\Gamma}\mathbf{\Lambda}^{-1}\mathbf{\Gamma}' = (\mathbf{X}'\mathbf{X})^{-1},$$

since $\mathbf{\Gamma}' = \mathbf{\Gamma}^{-1}$ and $\mathbf{\Lambda}^{-1}$ is a diagonal matrix.